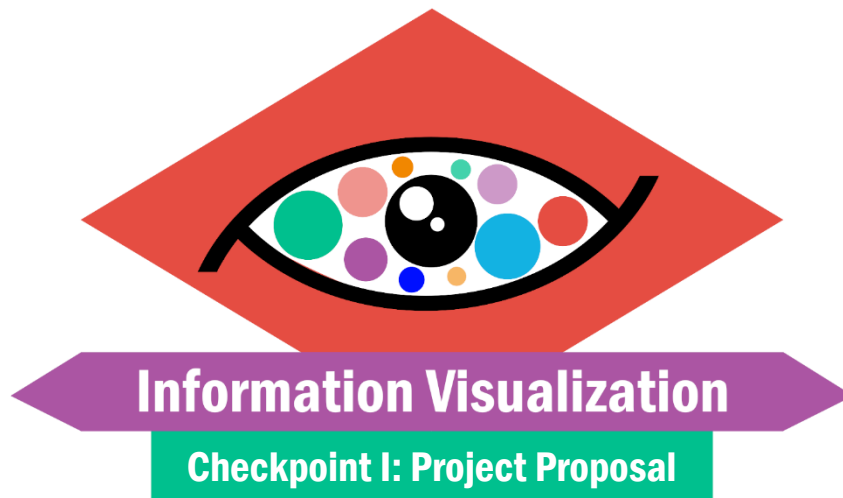




Objective

You must present the Task and Data Abstractions in Checkpoint I according to the What? Why? How? framework. Motivate why this domain is worth investigating, which insights you want to extract, and a convincing dataset exists for your visualization. At the end of this Checkpoint, you must have the tasks and data files for your visualization.

Requirements

Look at the course materials to understand the questions you need to answer.

Problem Domain

What is the domain you are going to tackle? Economy? Oscar winners? College student enrolment grades?

Choose a subject you find interesting and exciting. You must provide a **high-level description** to motivate the rest of the document. **Do not be too vague.** Avoid stating your project as a “visualization of COVID-19 world data”. Instead, it should be about “how COVID-19 changed European commute”. We want to know your scope.

Why (Task abstraction)

What tasks and what purpose?

Think about which tasks your visualization should help the users perform and which concrete questions it should give an answer to. You must provide a **set of at least five different, representative concrete questions** (and their description) organized in bullet points. We will look for the following question types: comparison, correlation, distribution, flow, geolocation (events, relationships, paths, or regions), hierarchy/inclusion, ordinal time, (cyclic/repeating) patterns, proportion, rank over time, relationships, time intervals, trends, and uncertainty. This does not mean you have to come up with a task of each of these types, but you must tell us which task type your tasks are.

Please provide a description that allows understanding the questions and their complexity. It will be reflected in your grade. Question enunciation must be as clear and concrete as possible.

For each question, **begin by writing its abstract form.** For example, the questions “Does age influence daily calorie intake?” and “Does the daily number of steps decrease with age?” both examine how one continuous variable (age) correlates with other continuous variables (daily calorie intake and daily number of steps). Their abstract form would be: “Does X decrease with Y?”

By identifying this abstract form first, you can **avoid creating questions that are too similar** or merely variations of the same underlying pattern. After defining the abstract form, you may proceed to write the specific version of your question.

If you are struggling to come up with a variety of tasks for a given scope, consider choosing a different one. Aim for diversity in both the types of tasks and their complexity. A set of trivial, easily answerable questions will not result in a strong project, and your grade will reflect that.

Keep in mind: you will be creating a dashboard that integrates multiple visualization techniques, integration is key. Do not treat tasks as isolated questions to be answered by separate, unrelated techniques. At least some tasks should emerge from the interaction between different components of the dashboard. A one-to-one mapping like “Technique A for Task A; Technique B for Task B” is not the right approach. Thoughtful planning for this integration will lead to a more interesting and complex final result. **So, guarantee that one task will be answered with more than one visual idiom.**

Beware that the tasks you choose now will be fixed for all remaining checkpoints. **If you are unable to answer one or more of your questions later, you will be penalized.** Plan carefully to ensure that everything proceeds smoothly.

What (Data abstraction)

Which dataset(s) and how will you process them?

Please tell us **which dataset you will use** and **where it is available**. Describe the dataset. Is it static or dynamic? How many variables are you dealing with? And how many items?

You can collect your data (e.g., using web scrapping or an API) if you want (it does not have to be a pre-made dataset). You can (probably should) also take data from several places instead of looking for this one website where you can download a zip file with exactly what you need. It may not exist and if it does it is an alarm sign: are you really looking for data to answer the questions that are relevant, or are you overfitting your questions to some “easy” dataset you found somewhere?

Still, remember you only have one week to prepare this so work smart, not hard, and do take advantage of existing datasets if they match your questions. Describe how, what sources, and what effort is involved to get **the initial dataset**, independently of where you got it from.

Parse the original data into .json or .csv format to use with D3. The initial dataset will have things in the wrong format. It will have attributes you do not care about. Some may have missing values or outliers. You may have two tables that you need to merge into one. You must describe how you processed the dataset (how you cleaned it, problems found and solutions, how you fixed missing values, cross-referenced different tables/datasets, etc.). If you do not need to process the data, explain why.

Describe which variables you selected from the dataset and which derived measured (**if needed**) you consider, computed, and why (based on your tasks and questions). Remember: you must visualize information and not data. Think about derived measures. If you did not derive any measure, carefully justify why you did not need any.

Having done all that, the main part of this section will, then, be the description of the final dataset that you will use in the next checkpoints/project, including:

- Description of the **dataset type** (spatial, table, field, etc.).

- Description of **each item** and **attribute** (nominal/ordinal/etc., diverging/sequential scale, etc.). Be precise. For instance, if you cover time-based variables, we want to know all features (linear/cyclic, hierarchic or not, spans, intervals, or instants, etc.).
- **Semantics** (what does each attribute and item stand for). We recommend using a table to save some space. Do not forget to include **all variables** (yes, including the derived measures).

As with the questions, the dataset you choose at this stage will be fixed for all remaining checkpoints. You will not be penalized for adding new attributes later if they improve your overall results. However, **you will be penalized if any of the attributes defined at this checkpoint are not used**. Make sure to include only the attributes you truly need.

Finally, although this may have been implied, we want to make it explicitly clear. After all data processing, you must end up with a single dataset. **Your dashboard must be powered by one unified dataset** to ensure smooth integration. This does not mean you are limited to a single file. It simply means that your dashboard backend should be able to handle one or more files at the same time to provide data to all visual idioms of the dashboard.

Mapping

Show **concrete** examples so that you can answer the questions you formulated with your data sample (“we want to know X, so if we look at attribute Y of table Z, and compare it with attribute W...”). Include **all questions** from the Task Abstraction. *No screenshots!* This is an important step for you (and us...) be sure that the data that you have is adequate for your goals.

Deliverables

Create a **4-page document using the provided template**. Follow the sections of the template but add more sections as you see fit. You should also **include in the submission** the **final dataset** that you are going to use in the project. **Compress everything in a .zip** and submit it online, until two days before your class (ex: classes on Monday must submit until Friday end of day).

In case you are having difficulties in selecting your data source and problem domain, check out the course pages on “Data Sources” and “Visualization Ideas”. **Notice that some visualization domains are banned, and you cannot use them.**

The way **NOT** to do this: looking up some vis examples (especially copy & pasteable ones...) and trying to find a subject that fits. Choose a subject you are passionate about. Choose a domain you are familiar with. Choose questions that have been keeping you awake at night. The result will be much better.

Example

You have already done something similar in the Wine Tasting Workshop. What you are doing here is very similar to our initial steps. First, we chose the domain and derived some interesting questions we wanted to answer. Then, we found the dataset. Remember the first question: Which year has the best points average? There was no attribute regarding the year. We needed to process the data and extract the “year” from the “title” attribute. We also had to clean the data. Most items had no “region_1” or a “region_2” unless they were from the USA. We also filtered the dataset to have tastings referring solely to Portuguese wines.

Another interesting question focused on whether there was a positive relationship between the wine’s price and rating. The question type is simple to guess: correlation. We have two quantitative variables (the price is a ratio, and the rating is ordinal), and we are trying to find out if there is a linear relationship between them. How can we show that we have the data to answer the question? The columns “price” and “points” have the data needed for this specific question:

```
(from "dataset_wines.csv")
```

```
id,points,price,region, (...)  
70163,89,50,Douro, (...)
```

These are some decisions and jobs you must make at this checkpoint. However, these are just broad strokes. We want you to explain everything more in-depth.

Penalties

- Documents over 4 pages long: **1 grade point penalty per extra page.**
- Document uploaded after the deadline: **0.5 grade points penalty per hour of delay.**
- Document template altered (wider margins, smaller font, etc.): **1 grade point penalty.**
- Missing data files: **1 grade points penalty.**
- Incorrect file extension: **0.5 grade points per file.**

Tasks to perform during the lab

The professor will provide feedback. The grade will be made known one week later (see below). If you are not receiving feedback, then you must be peer assessing your colleagues' submissions.

Grading

Your proposal will be graded according to the following parameters:

- **Theme registration in Moodle (5%):** you must register a valid theme in Moodle.
- **Domain identification and motivation (10%):** clearly identify the theme you tackle and justify why it is interesting and relevant.
- **Task abstraction (35%):** present at least five different questions, identifying their type and given concrete examples. Explain the rationale behind each question. Diversity of types and complexities will be taken into consideration.
- **Data abstraction (35%):** present and describe the initial dataset, the data parsing/processing, and the selected and derived measures.
- **Mapping (10%):** show that you can answer each question with data samples.
- **Peer assessment (5%):** grade your peers' submissions in the class.

An important note on grading for this and all other Checkpoints: always *justify* your choices, based on the basic principles you have learned in this course (adequacy of channels to data types, human perception, etc.). Do not just describe what it is, but especially *why* it is as it is. **Common sense and ChatGPT will not save you here.**

Additional Notes

After you deliver your document, your work will be graded. HOWEVER, this grade **can be improved by up to two grade points** if you correct any faults pointed out by the professor and submit a revised version of the document **HIGHLIGHTING THOSE CHANGES** before the beginning of the class taking place 7 days after you receive feedback in class. **Only highlighted changes will be considered.**